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Review

Machine learning-based detection of transportation infrastructure internal defects using ground-penetrating radar: a state-of-the-art review

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Abstract

Early detection of internal defects is crucial to ensure the long-term performance and stability of transportation infrastructure. Researchers and practitioners have developed various nondestructive testing (NDT) methods for this purpose. Among them, the ground-penetrating radar (GPR) technique has been widely implemented due to its advantages such as large coverage, traffic-speed survey, and rich subsurface information. In addition, machine learning (ML) algorithms have been frequently applied to achieve automatic GPR data interpretations, which are essential for field applications. However, the fundamental concepts, architectures, and appropriate application scenarios of these algorithms are often questionable to practitioners and researchers. This paper presents a state-of-the-art review of ML applications in the internal defect detection of transportation infrastructure using GPR. In particular, pavements and bridges are covered. The basic knowledge of GPR working principles and ML algorithms is documented. The critical features of the ML algorithms for each detection task are presented. The drawbacks that may hinder the application of ML algorithms using GPR are indicated, including the insufficiency of labeled GPR data, unavailability of GPR dataset, customized ML architecture, and field validation. Finally, possible transfer learning, integrated robotic platform, and data fusion with other NDT methods are discussed. This review paper is expected to serve as a reference for practitioners to choose appropriate ML algorithms to detect internal defects in transportation infrastructure using GPR.

Keywords: transportation infrastructure, machine learning, ground-penetrating radar, pavement, bridge, internal defect

1. Introduction

Transportation infrastructure, such as highways and bridges, may suffer from internal defects during service, which may develop into severe deterioration and cause safety issues. This necessitates routine and effective inspections to monitor the internal conditions of transportation infrastructure. Both destructive and nondestructive testing (NDT) techniques can be applied for this purpose. Destructive methods, such as drilling or excavation, can provide accurate subsurface conditions. However, they could jeopardize the homogeneity or strength of the structure. Inappropriate fixing or patching of the testing locations may also result in future damages. Furthermore, destructive tests can only be performed at limited locations, which cannot reveal the full picture of the subsurface conditions.

To overcome the limitations of destructive methods, various NDT methods have been developed and implemented for assessing the subsurface condition of transportation infrastructure (Lytton 1990; Gucunski and National Research Council 2013). These methods can be divided into passive methods and active methods. Passive methods receive mechanical or electromagnetic (EM) signals from infrastructure at internal defect locations, while active methods send signals and analyze the internal conditions based on the reflected signals (McCann and Forde 2001). Examples of conventional NDT methods for surface and/or subsurface condition assessment include visual inspection, hammer sounding, chain dragging, impact echo, ultrasonic, infrared thermography, and ground-penetrating radar (GPR). Among these methods, GPR has been widely used for subsurface condition investigation as a powerful and versatile tool (Lahouar 2003). It is based on EM wave propagation and reflections (Annan 2005). It may allow traffic-speed data collection to avoid traffic congestion, which is extremely important for highways and bridges with high traffic volumes. It can also enable large-coverage, non-contact, and continuous data collection. However, GPR measurements may be vulnerable to high-conductivity materials, such as wet clays and heterogeneous environments, that may cause signal attenuation and EM wave scattering. The raw data collected from GPR devices need to be interpreted to unveil the actual subsurface conditions, which could be non-intuitive and require considerable expertise.

Appropriate processing of GPR data is essential for accurate subsurface mapping, which also depends on the GPR equipment. Over the years, various signal processing and data interpretation methods have been developed and implemented. Conventional

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signal processing methods based on EM wave propagation theory were first developed (Benedetto and Pajewski 2015; Daniels 2004). These methods include amplitude and phase analysis, digital filtering, deconvolution, migrations/inversions, and image analysis. However, these methods may require significant knowledge of EM wave theory. In addition, the prediction accuracy highly depends on the processing methods and parameters.

Machine learning (ML) algorithms based on 1D signals and 2D images have been developed to identify the patterns caused by internal defects (Giannakis et al., 2019). Feature extractors combined with conventional signal processing methods have been used to reduce noise and increase the signal-to-noise ratio (SNR) of the target patterns. Then, nonlinear mapping algorithms, such as neural networks and support vector machines (SVMs), may be implemented to achieve pattern recognition, location, feature point regression, and segmentation of localized subsurface defects (Ozkaya and Seyfi 2018; Harkat et al., 2019). In addition, feature points and segmentation results may allow the reconstruction of the buried objects in three dimensions (Liu et al., 2021a). These ML or deep learning (DL) methods have dominated GPR data interpretation. However, despite the high detection accuracy and efficiency reported in the literature, the principles of these algorithms are often treated as black boxes, which require interdisciplinary knowledge for interpretation. In addition, choosing appropriate algorithms for a specific detection job is usually challenging as some have comparable performances, where advantages/drawbacks often need to be identified by users.

This paper provides a comprehensive review of the existing ML algorithms that have been successfully implemented for detecting the internal defects of transportation infrastructure. First, the GPR principle is briefly introduced in Section 2. Then, the fundamental ML and DL concepts and algorithms are briefly introduced, followed by ML- and DL-based GPR applications in detecting transportation infrastructure internal defects in Section 3. The discussion on the existing problems and future trends and conclusions are presented in Sections 4 and 5, respectively.

2. GPR working principle

GPR is an NDT tool based on EM wave propagation and reflection in the subsurface domain. EM signals are generated and sent from the transmitting antenna. Materials with a variation in dielectric constant can generate reflection signals, which can be picked up by the receiving antennas. The transmitting and receiving antennas can be in the same or two separate shielded boxes. The intensity of the reflection signal is proportional to the dielectric constant difference between the two materials (Davis and Annan 1989). Subsurface conditions can be reconstructed by the signal analysis of the transmitted and reflected signal waveforms and frequency spectra.

During data collection, EM waves are transmitted and received beneath the antenna along the depth direction of the surveyed infrastructure. This generates a 1D signal called trace or A-scan, which has a time duration in the nanosecond scale. These traces are generated at a specific frequency as the antenna moves, which is referred to as the spatial sampling frequency. Hence, a 2D image with y-axis of EM wave propagation time and x-axis of surveyed distance can be obtained, which is referred to as a radargram or B-scan. By estimating the time-of-flight and propagation speed of the EM wave, the location of a specific reflector can be determined (Jol 2008). This is the principle of internal defect detection using GPR.

Both pulse and stepped frequency antennas have been used in GPR applications (Pajewski et al., 2015). A pulse antenna has a central frequency with a wide frequency bandwidth, which enables short wavelength in the time domain to obtain high spatial resolution based on the Fourier transform. This is called an ultra-high-frequency radar system. Usually, high-frequency pulse antennas are suggested for shallow and small-scale defect detection due to the higher horizontal and vertical resolutions compared with low-frequency antennas. However, this could lead to limited penetration depth as signal attenuation is proportional to the frequency.

The stepped frequency antenna generates EM waves in the frequency domain (Suksmono et al., 2010). The lower limit frequency is produced first and then increased at specific steps until the higher limit. This antenna type may reach both deep penetration range and resolution (Langman and Inggs 2001). However, it has a more complicated waveform than the pulse antenna. The pulse antennas used in transportation infrastructure applications can be generally divided into ground-coupled and air-coupled models (Wang et al., 2022). The ground-coupled one requires the antenna to be attached to the pavement/bridge/wall surface to ensure the penetration of EM waves. This increases penetration depth at the cost of relatively slow survey speed, which can be addressed by the air-coupled antenna. For example, traffic speed surveys on highways and bridges are often conducted using air-coupled horn antennas with high central frequency.

Conventional signal processing is the key to GPR data interpretation. The post-processing methods for GPR applications have been studied for decades and can be divided into trace-based and image-based categories. The trace-based methods usually include digital filtering, time-varying gain, and deconvolution to suppress the noise and increase the SNR of the target signal features (Wang et al., 2020). Image-based methods mainly deal with B-scan images. Many computer vision approaches have been implemented to enhance the reflection features of targets in the GPR data. Various migration approaches have been developed to unveil the accurate geometric information of subsurface objects.

3. ML-based GPR applications in defect detection

ML-based techniques to extract defects from GPR data have been widely implemented in transportation engineering. In this section, ML concepts and algorithms are reviewed. A state-of-the-art survey of ML-based GPR applications in transportation infrastructure, mainly pavement and bridge, was conducted.

3.1 ML background

3.1.1 *Machine learning*

ML is a pivotal part of artificial intelligence (AI) techniques, featured by shallow architectures. It allows the computer to learn itself from a booming amount of data without being explicitly programmed. The primary classification of ML and corresponding algorithms under each category are presented in Fig. 1 (Thai 2022). Among them, supervised learning algorithms relying on labeled data are dominantly applied in civil engineering. The first step is to acquire raw data consisting of a pair of variables: feature (input) and label (output). Then, the raw data will be randomly divided into training and testing sets. In some models, a third validation set is split from the training set to evaluate the trained model during hyperparameter fine-tuning (Yang and Shami 2020). In the second step, the selected goal-orientated algorithm will be trained

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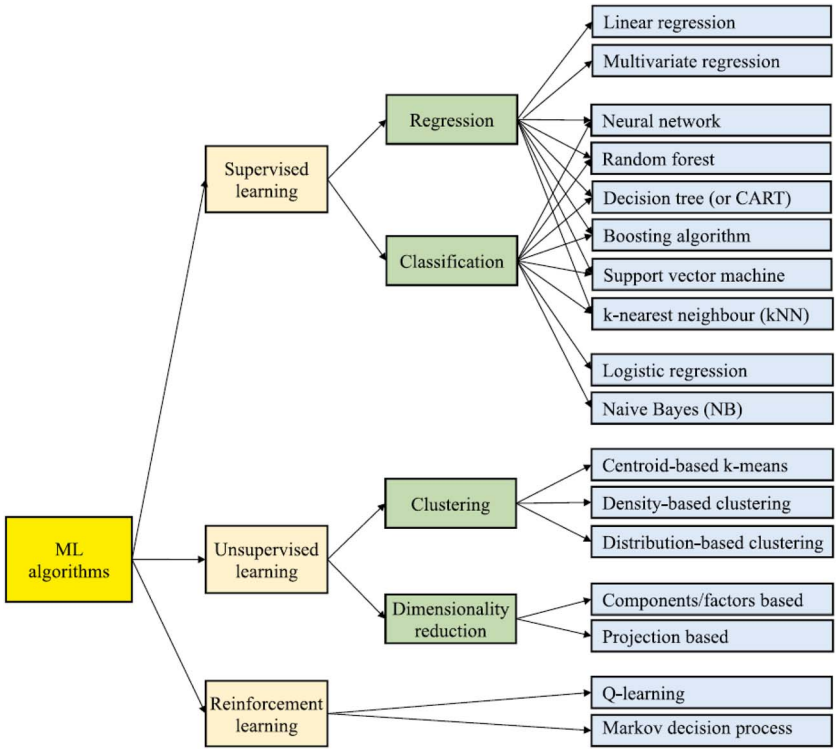


Figure 1. Classification of ML and corresponding algorithms (Thai, 2022).

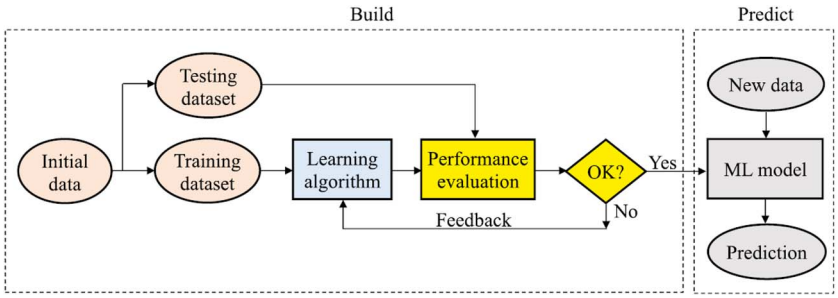


Figure 2. Typical workflow of ML (Thai, 2022).

to build an ML model whose performance will be evaluated with the testing set by using linear or non-linear loss function as the indicator (Mahesh 2020). The workflow of generic supervised learning is illustrated in Fig. 2 (Thai 2022).

Fast prediction speed and accurate results boost ML's widespread applications in regression and classification tasks using GPR data. Civil engineering scenarios include designing and fine-tuning antennas, predicting material property and classification, detecting buried objects, and detecting defects and deterioration of infrastructure. Algorithms that have been implemented include support vector machines (El-Mahallawy and Hashim, 2013; Xie et al., 2013), regression analysis (Giannakis et al., 2020; Morris et al., 2020), decision tree (Syambas, 2012), random forest (Zhang et al., 2018), neural networks (Shaw et al., 2005), genetic algorithm (Harkat et al., 2019; Mohammed Abdelkader et al., 2019; Tang et al., 2021), Viola-Jones object detection framework (Maas and Schmalzl, 2013), Naïve Bayes (NB) (Gibb and La, 2016), and Histogram of Oriented Gradients (HOG) (Kaur et al., 2015). However, several drawbacks hinder ML's implementation in the GPR community, i.e. the shallow architecture of the algorithms (few or only one hidden layer), the requirement of a carefully designed feature extractor (LeCun et al., 2015), and insufficient GPR data or open datasets in early years. The usage of ML is mainly debated and explored within an academic domain, while its application at the project level is still limited. At the same time, more sophisticated DL algorithms that could deal with millions of parameters have gained attention. Their ability to automatically extract targeted features from background clutter in GPR images made them preferable in various domains, including transportation engineering (Tong et al., 2020a).

3.1.2 *Deep learning*

DL, a branch of ML with deep hierarchical architectures consisting of multiple non-linear stages that can solve more complex representation tasks with sufficient data and tremendously increased computing capacity, shows superiority over conventional ML algorithms (Krizhevsky et al., 2017). According to Deng (Deng, 2014), DL algorithms can be classified into generative architectures, such as autoencoder and deep belief neural, discriminative architectures, such as convolutional neural network (CNN) and recurrent neural network, and hybrid architectures. All these architectures have been reported to successfully interpret GPR data (Alvarez and Kodagoda, 2018; Ozkaya and Seyfi, 2018; Picetti et al., 2018).

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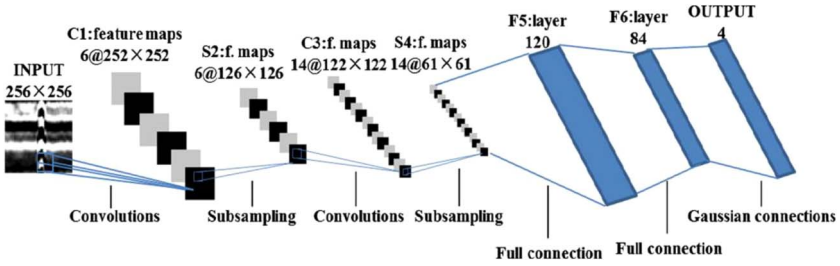


Figure 3. The architecture of a generic CNN (Tong et al., 2020a).

Due to the limited information from the 1D GPR signal and the high cost of purchasing or custom-manufactured 3D/array radar, the 2D B-scan images are still preferred by practitioners in field applications. This leads to prosperous endeavors of applying multiple CNN-based object detectors upon subsurface defect detection tasks among GPR communities. CNN, namely, ConvNet, has been successfully implemented in object detection and image segmentation (Girshick et al., 2014; Szegedy et al., 2015; Tompson et al., 2015). Typically, a generic CNN architecture has multiple stacked convolutional layers and pooling layers, as shown in Fig. 3 (Tong et al., 2020a). These two unique modules were used to replace part of the fully connected layers (FCNs) in conventional deep neural networks, resulting in better generalization performance (LeCun et al., 1989; LeCun et al., 1998). Based on the architecture's complexity, these object detectors can generally be classified into two- and single-stage object detectors. The former category includes region-based convolutional neural network (R-CNN) and its successors, i.e. Fast R-CNN, Faster R-CNN, and feature pyramid network (FPN) (Girshick et al., 2014; Girshick, 2015; Ren et al., 2015; Lin et al., 2017). This object detector proposes the region of interest in the first stage, and the bounding box is then located in the second stage. The detection performance of the model was usually evaluated by metrics such as the confusion matrix containing precision, recall, average precision (AP), mean average precision (mAP), F1-score, receiver operator characteristic (ROC) curve, and inference time (Davis and Goadrich, 2006; Powers, 2020). High accuracy was reported applying this kind of CNN-based model upon GPR B-scan (Pham and Lefèvre, 2018; Xu et al., 2018).

Another prevailing type of CNN-based object detector is the single-stage object detector, represented by you look only once (YOLO) models. The original YOLO v1 (Redmon et al., 2016), inspired by GoogleNet (Szegedy et al., 2015), identifies and locates the objects in one stage, reframing the detection task as a regression process. A booming development of the YOLO series and successors have been developed in recent years, including YOLO v2 (Redmon and Farhadi, 2017) inspired by VGG (Simonyan and Zisserman, 2014) and network in network (NIN) (Lin et al., 2013), YOLO v3 (Redmon and Farhadi, 2018) inspired by FPN, YOLO v4 (Bochkovskiy et al., 2020), etc. The detailed concepts, architecture, and techniques adopted by YOLOs could be found in (Diwan et al., 2023). The feature extractions in GPR images using YOLOs show remarkable robustness in the studies by different research groups (Li et al., 2020; Park et al., 2021; Liu et al., 2021b), which indicate that it is a promising tool for GPR data interpretation.

3.2 Pavement

Pavement is an essential component of modern transportation infrastructure. It can be generally categorized into highway pavement and airport pavement in terms of load type. Based on the structural design and material properties, it could be primarily categorized into flexible, rigid, and composite pavement. The serviceability and life of such layered infrastructure are affected by material, structural design, and periodic maintenance. The optimum performance is expected during the prolonged pavement service life, which requires a time- and resource-conserving inspection of the pavement's health condition. With the merit of the detectability of subsurface distresses, GPR has gained attention of various departments and transportation agencies. Van- or unmanned aerial vehicle (UAV)-mounted air-coupled GPR systems are preferred in pavement condition inspection to achieve high-speed survey without traffic closure. Automatic subsurface pavement distress recognition is crucial because of the time intensity and labor intensity of the manual interpretation of massive GPR data collected by different agencies and researchers. Advanced AI techniques, such as ML- and DL-based algorithms, emerged and have been adopted as reliable solutions. In this subsection, state-of-the-art reviews about ML- and DL-based defect inspection in highway pavement and airport pavement are provided.

3.2.1 *Highway pavement and urban road*

Due to its riding comfort, flexible pavement is more frequently used in road and pavement construction, especially for high-level highways. Inevitably, various distress types appear under the coupled effect of vehicular loading and environmental factors such as temperature and moisture. According to the multilayer theory, the mechanism behind these distress types can attribute to excessive strains in different layers. For health monitoring purposes, these defects can be divided into four categories: cracking, viscoplastic deformations, surface defects, and miscellaneous distress (Ragnoli et al., 2018), which are discussed in the following text using ML and DL methods.

Cracking

Cracking is a predominant distress in highway pavement systems, undermining the riding quality and structural integrity. Surface cracks could be detected and localized by visual inspection, digital camera, infrared thermography, and light detection and ranging. However, they cannot detect subsurface concealed cracks (Solla et al., 2014). Early identification of the severity of concealed cracks contributes to decision-making in QC/QA procedures, which is ecofriendly and financially friendly for maintenance and rehabilitation purposes. GPR relies on the propagation of EM waves within the layered pavement system. Position and geometry information of concealed cracks could be captured by the hyperbolic features in the GPR radargram (B-scan or C-scan) (Diamanti and Redman, 2012; Krysiński and Sudyka, 2013). However, detection accuracy remains unsatisfactory using traditional velocity analysis methods due to the limited vertical and horizontal resolutions of GPR

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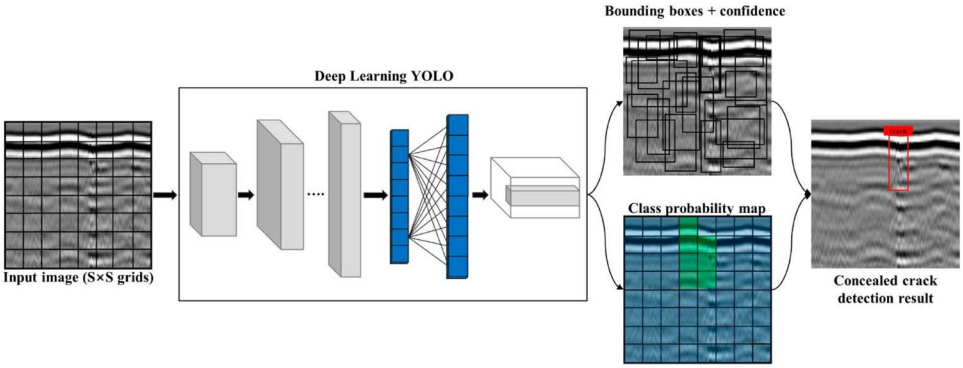


Figure 4. GPR B-scan image-based concealed crack detection in asphalt pavement using YOLO models (Liu et al., 2021b).

and field noise in real applications. Approaches, including preprocessing, cluster removal, segmentation, and feature extraction, were mainly applied to the GPR data interpretation procedure to enhance the detectability of concealed cracks (Torbaghan et al., 2020).

ML and DL algorithms were developed to recognize concealed cracks automatically. YOLO models show superiority on this task with high efficiency and accuracy. Liu et al. (2021b, 2022) compared and evaluated the performances of several YOLO models (mainly v3, v5, and their variants) targeting concealed cracks and voids upon 3D GPR data, as seen in Fig. 4. The results showed that examined models could achieve 80∼94 per cent mAP. Economic and environmental benefits could also be obtained compared with traditional inspection methods. In their studies, the hyperparameter optimization (HPO) method using Bayesian search was performed and verified to improve accuracy. The performance of the YOLO series on concealed crack detection was also examined by another study, which makes this type of objective detection model an effective tool for interpreting GPR images (Li et al., 2021a).

Moisture damage

Asphalt pavement usually suffers from moisture damage induced by dynamic pore pressure (Wang et al., 2019). The flushing and pumping caused by moisture weakens the cohesion between aggregate and asphalt binder. Moisture damage at the pavement surface mainly includes stripping, raveling, and pothole, which can be observed by visual inspection. NDT methods, including GPR, could identify the damage caused by infiltrated moisture inside the base layer or subgrade. According to EM mixing theory, the dielectric constant of the asphalt mixture can be determined by each constituent's electric and volumetric properties. Due to the high dielectric constant of water, which is 81, GPR is extremely sensitive to moisture variation in the porous medium. The influence of moisture could also be camouflaged by other factors that affect the bulk dielectric constant of asphalt mixture, such as density, compaction degree, layer thickness, gradation, etc. Traditional velocity or time-frequency analysis methods could barely isolate moisture's effect among these coupled factors.

Meanwhile, the spatial distribution of moisture in the base layer or subgrade depends on the interconnected pore structure, which could be randomly distributed. Thus, identifying and locating moisture damage directly through GPR B-scans requires rich experience. DL-based methods, instead, extract morphological features from radargram, which solves the issue by preventing prediction of the dielectric constant of asphalt mixture and individual effort on pattern searching. A CNN YOLO-ResNet50 model was proposed by Zhang et al. (2020a), aiming at the fast and accurate automatic detection of moisture damage through GPR B-scan, as seen in Fig. 5. The deep ResNet50 base network serves as a feature extractor in this model. YOLO v2 was added as an object detector frame to generate the bounding boxes. To train and fine-tune the CNN model with the moisture damage dataset, *K*-means clusters were applied, replacing the original anchor boxes in the YOLO v2 algorithm. Inspired by the particle filter method, an incremental random sampling (IRS) method was proposed to address the low contrast issue between the target (moisture damage) and background.

Subgrade defects

Subgrade is the foundation of highway structure, which provides load-bearing capacity. Precipitation from pavement seepage and water table leads to subgrade defects such as uneven settlement and cracks. These subgrade defects could propagate upward and induce further surface and base layer distress. GPR shows the potential for subgrade defect detection using the DL-based algorithm despite the effects of noise and background clutter. Tong et al. (2018) examined the performance of multi-stage and cascade CNNs to detect uneven settlement, sinkhole, and subgrade cracks, as seen in Fig. 6. Similar studies with other algorithms like Sobel edge detection and *K*-value clustering analysis also showed the superiority of the proposed CNN-based models. The cascade CNN model achieved the highest recognition accuracy (97.35 per cent) among all examined ones on subgrade defect detection. In another work conducted by Tong et al. (2020b), the NIN model was utilized to detect pavement distresses such as cracks, water damage pits, and uneven settlement. In this model, multilayer perceptron (MLP) layers were used to replace traditional pooling layers to increase accuracy, which has been verified by a previous study (Springenberg et al., 2014). Another feature of this NIN model is that the global average pooling layer was used instead of FCNs, avoiding overfitting and poor generalization performance. The results showed that the proposed NIN model overperforms the traditional Faster R-CNN and SVM models in terms of accuracy and average loss. However, it should be noted that the reported accuracy on subgrade defects relies on the accurate label of the defects and model structure, which makes it experience-dependent at the project level. For example, using a different dataset, the Faster R-CNN algorithm was reported with higher accuracy for the same targeted defects as Tong et al. (2020b).

The underground cavity is a critical subgrade defect that requires early detection due to its severity to property safety, especially in urban roads and pavements. Heavy precipitation,

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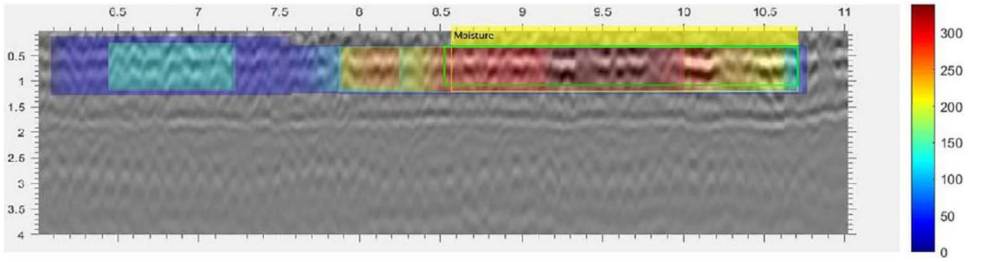


Figure 5. Heatmap of the moisture damage detection result of GPR data by IRS method and YOLO-ResNet50 model (Zhang et al., 2020a).

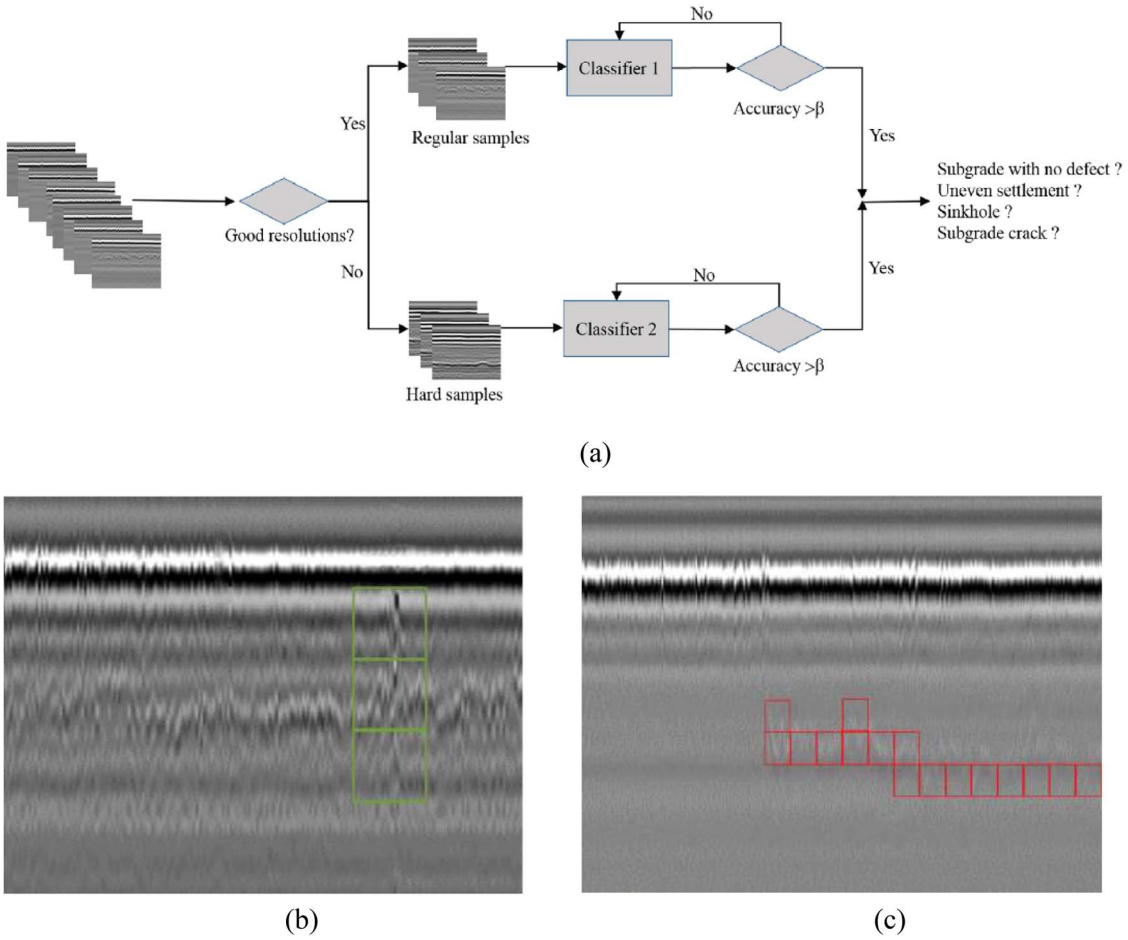
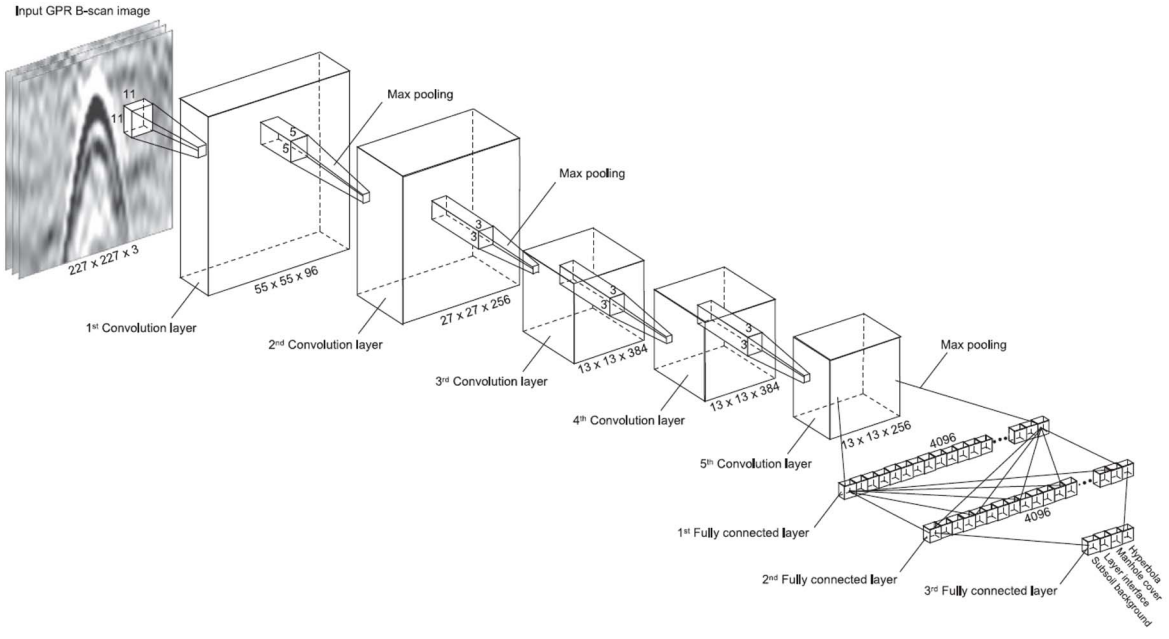


Figure 6. Subgrade defect detection using cascade CNN: (a) structure of the cascade CNN; (b) detection results of subgrade crack; (c) detection results of uneven settlement (Tong et al., 2018).

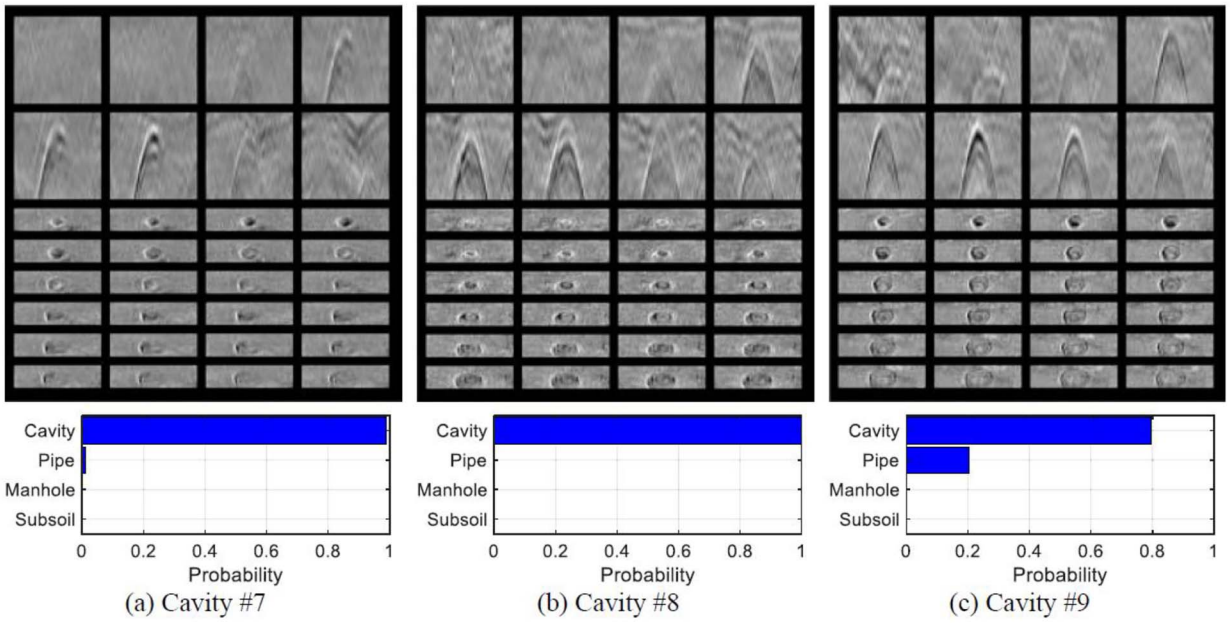
leakage of water pipes, and collapse of underground utilities or caves could lead to such defects (Liu et al., 2021a). In addition, cavity morphology is a key factor that affects the severity of pavement catastrophes, which needs to be detected and characterized using GPR. GPR B-scan alone or with C-scan is preferred to quantitatively determine the cavity morphology (Hou et al., 2022a). Usually, ground-coupled antennas with low central frequency are preferred because of the larger penetrating depth. Detecting road pavement cavities requires intensive labor and sources, which necessities automatic recognition through ML- or DL-based approaches. Qin and Huang (Qin and Huang, 2016) applied SVM with feature vectors composed of discrete cosine transform (DCT) and polarization attributes in detecting underground voids through the finite integration technique (FIT). The proposed method is robust in separating voids from other objects, such as concrete, mild steel, and medium-density polyethylene, under different noise sources. Acquisition of on-site GPR cavity data is labor-intensive. To address the minor dataset issue, Hou et al. (2022a) employed a relation network (RelationNet) and a few-shot learning (FSL) framework to classify cavities in different shapes, including spherical, rectangular, cylindrical, and hemispherical, without scarifying detection accuracy and efficiency. Other CNN-based methods were also reported to be applied in underground cavity detection (Kang et al., 2019; Khudoyarov et al., 2020; Kim et al., 2020, 2021) and object detectors, such as YOLO v3 (Ni et al., 2020). Figure 7 shows an example of DL's application in underground cavity detection (Kim et al., 2021).

DL-based models achieved high detection accuracy on the main concerned pavement distresses. However, false detections

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(a)



(b)

Figure 7. Underground cavity detection using 3D GPR data and CNN: (a) structure of the used CNN; (b) detection results of the underground cavity using the proposed method (Kim et al., 2021).

on objects such as subsurface voids were also reported using such models when highly reflective surface objects exist (Yamaguchi et al., 2018). Yamaguchi and Mizutani (2021) developed an SVM model based on 350-km 3D GPR road survey data to automatically classify and localize manholes and joint covers. The HOG feature was applied to extract the geometric characteristics of manholes and joints. The Laplacian filter was utilized as a preprocessing filter to enhance the object edges. Evaluated by the confusion matrix, the manholes and joints detection algorithm showed a 98 per cent classification accuracy. Hough transform, a signal-based processing method, was embedded in the algorithm for incorporating the detected objects into maps. Note that this method's accuracy is operator-dependent, requiring known object sizes and orientation in advance (Li et al., 2016).

In subgrade defect detections, the detection accuracy of ML/DL-based subgrade defect detection using GPR depends on both the detecting resolution and the scale of defect itself. GPR with lower frequency is usually adopted for this scenario, resulting in a relatively lower resolution. However, the larger geometric scale of subgrade defects, such as cavity and

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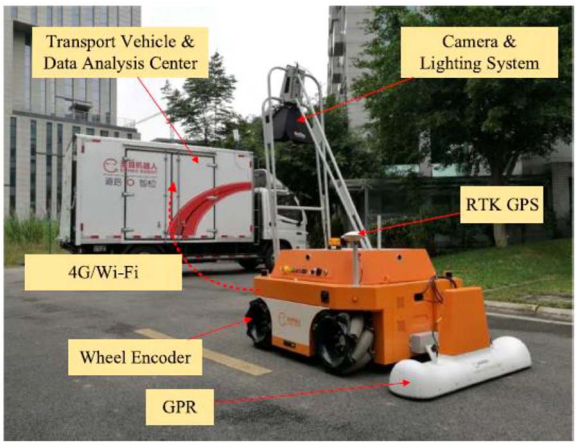


Figure 8. Robotic system for airport runway defect inspection used by Li *et al.* (Li et al., 2021b).

underground utility, make their waveform distinguishable to their surroundings. Therefore, comparative detection accuracy could be achieved using the YOLO series or CNN-based models.

3.2.2 *Airport pavement*

Airport pavement structures include runways, taxiways, and apron pavements. High contact pressure caused by airplane taking-off and landing results in subsurface defects, including voids, cracks, debonding, etc., requiring routine inspections. NDT methods, such as GPR, heavy weight deflectometer, and impact echo, are preferred due to the limited inspection period and safety requirements (Solla et al., 2021). Intelligent inspection platforms and advanced data fusion techniques are frequently applied, which provide the advantages of high inspection accuracy and fast inspection speed.

Advanced DL algorithms were applied for data interpretation purposes. Li et al. (2021b) developed a GPR-RCNN algorithm using the 3D GPR data from a runway inspection robotic system, as seen in Fig. 8. The proposed RCNN-based algorithm is a two-stage framework that can extract complex features of subsurface defects. In the first 3D proposal generation stage, GPR C-scans were subjected to the semantic segmentation in a bottom-up manner. Voxel-wise and planar features were extracted through a sub-network consisting of 3D/2D convolution, BN layer, and ReLU layers sequentially. Then, the obtained features were fused to fulfill the foreground segmentation, followed by 3D box proposals. In the second 3D bounding box refinement stage, the box locations and orientations were refined through region pooling and three FCNs. It should be noted that the performance of the proposed GPR-RCNN algorithm varies between different runway defects or objects in the literature. For example, an 87% F1-measure could be achieved for pipes detection task. However, this value reduced to 33% for crack detection task. In the study by Li et al. (2022a), a DL algorithm, named MV-GPRNet, combining 3D and 2D CNNs simultaneously, was further developed. The corresponding F1-measure was improved to reach 69 and 100 per cent for cracks and pipe detection, respectively. It should be noted that crack identification remains challenging despite the improved performance of the DL algorithm via different model architectures. If the width of concealed cracks is comparable to or smaller than the EM wavelength, hyperbolic reflection features could be weak and difficult to detect.

3.2.3 *Summary*

In summary, ML and DL algorithms have become blooming trends for detecting defects in expressway pavement and airport runways using GPR. Concealed cracks, underground cavities and voids, moisture damage, subgrade defects, uneven settlement, manhole, and joint covers are task of interests. SVM, Hough transform, YOLO series, multi-stage CNN, cascade CNN, and NIN have been implemented by different researchers. Compared with traditional velocity analysis, or amplitude analysis, ML- or DL-based methods could automatically detect defects with higher accuracy. A summarization of current ML and DL applications in expressway pavement and airport runways defect detection using GPR data can be seen in Table 1.

3.3 Bridge

Bridge deck inspection is crucial to ensure the bridge structure's integrity and functions. Early detection of structural deficiencies in bridges could avoid potential accidents. GPR may provide rapid and safe detection without traffic congestion. GPR surveys can be carried out on concrete decks, girders, piers, or columns, depending on the target defect types. GPR has achieved high accuracy in the concrete bridge deck defect detection, such as rebar corrosion, cracking, spalling, delamination, and moisture damage, as well as structural assessment tasks, such as concrete cover depth estimation, overlap thickness measuring, and rebar localization and mapping. GPR data interpretation techniques in bridge decks can be borrowed from those applied in pavements due to the shared principles. For example, experienced operators are required to manually extract hyperbolic features from a field survey GPR B-scan (Solla et al., 2021). ML- and DL-based algorithms became the mainstream tools to interpret GPR data for concrete bridge deck inspection. This section summarizes the documented applications using computer-aided algorithms on bridge inspection in terms of different types of distress.

3.3.1 *Rebar localization, mapping, and corrosion assessment*

Assessments of reinforcement in bridge decks, such as rebars, pre-stressed tendons, and ducts, are the main tasks in bridge inspection. Precise detection of embedded rebars is beyond the detectability of surface methods, such as visual inspection. The rebar localization and mapping benefit the bridge structure in terms of maintenance/rehabilitation (Gucunski and National Research Council, 2013; Alani et al., 2014). GPR has shown great potential to assess rebar conditions because EM wave is highly sensitive to metallic material or objects. 3D rebar contour maps could be obtained using array GPR antennas or 3D antennas (Dinh et al., 2019). The critical step in rebar detection from bridge GPR B-scan is to capture the hyperbolic feature. This process is time-intensive if rebars are identified manually. Although image-based automatic recognition algorithms were proposed, the near-real-time monitoring of rebar conditions remains challenging. Low resolution and signal overlapping limit GPR's detectability if the reinforcement mesh/bars are densely distributed.

Advanced ML- and DL-based models were proposed recently to promote the accuracy and efficiency of bridge rebar detection. An ML-based model combined the HOG feature descriptors, and a supervised binary data cascade classifier, called AdaBoost, was proposed to automatically extract rebar location through GPR data (Asadi et al., 2019, 2020). In this supervised model, the classifier was trained offline during the processing and quantification of the bridge deck GPR B-scan. The performance of

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Table 1. ML/DL applications upon expressway pavement and airport runways defect detection using GPR data.

Tasks	References	Algorithms and methods	Data source and scale	Performance metrics	Features
Manhole and joint covers	Yamaguchi and Mizutani (2021)	SVM and Hough transform	SF-CW 3D radar; Manhole: 6300 Joint: 2200	Confusion matrix (precision, recall, F1 score)	High accuracy was achieved by the SVM model utilizing the Laplacian filter for preprocessing, the HOG feature for characteristics extraction, and Hough transform (visualization).
Concealed cracks and void	Liu et al. (2021a, 2021b) Liu et al. (2022)	YOLO v3 YOLO v5	SF-CW 3D radar; Concealed cracks: 1400	Confusion matrix (precision, recall, F1 score, mAP, FPS, ROC, AUC)	The YOLO series (mainly v3, v5, and corresponding variants) model was utilized. Economic- and environmental-benefits analyses were conducted. HPO was conducted using the Bayesian search.
Concealed cracks	Li et al. (2021a)	YOLO v3 YOLO v4 YOLO v5	3D Radar 303 images (B-, C-, and D-scans) containing 1,306 concealed cracks	Confusion matrix (precision, recall, F1 score, mAP, FPS) and inference time	YOLO v3, v4, and v5 models are applied to detect concealed cracks from 3D GPR data. Among these models, YOLO v4 has the best performance.
Subgrade defects	Tong et al. (2018)	Multi-stage CNN and cascade CNN	2D Pulse radar; uneven settlement and sinkhole and subgrade cracks: 312	Confusion matrix (accuracy)	Multi-stage CNN and cascade CNN models were developed. Sensitivity analysis in terms of transmitting frequencies and pavement structure.
Concealed cracks and uneven settlement and water damage pit	Tong et al. (2020b)	NIN	2D Pulse Radar: 1500	Confusion matrix (accuracy, average loss)	NIN models with MLP layers were employed and showed higher classification accuracy.
Airport runway defects and utilities	Li et al. (2021b) Li et al. (2022b)	GPR-RCNN MV-GPRNet	3D Radar: Cracks: 248 Voids: 20,605 Subsidence: 606 Pipe: 7467	Confusion matrix (precision, recall, F1-score)	A CNN-based two-stage framework, namely, GPR-RCNN, was proposed to detect and differentiate different defects and utility pipes in airport runway.
Moisture damage	Zhang et al. (2020a)	CNN and IRS method	2D Radar: Moisture damage: 2,135	Confusion matrix (precision, recall, F1-score)	Deep ResNet50 base network was used as feature extractor; YOLO v2 was added as object detector; IRS algorithm was proposed to detect moisture damage from raw GPR data automatically.
Underground cavity	Hou et al. (2022a)	RelationNet (a few-short learning framework)	3D FDTD simulation Cavity: 68	Confusion matrix (precision, recall, F1-score)	A RelationNet, a FSL-based framework, was applied, aiming to obtain acceptable accuracy and efficiency using a limited number of GPR images.
Underground cavity	Kim et al. (2020); Kim et al. (2021); Khudoyarov et al. (2020);	CNN method: AlexNet	3D Radar Cavity: 10,296 B-scan images (2018); 16 (2019)	Confusion matrix (precision, recall, F1-score, accuracy)	CNN-based algorithms were applied using B-scan and C-scan, separately or conjunctively.
Underground cavity Manhole Pipe Intact	Kang et al. (2019) Kang et al. (2019)	CNN method: UcNet Deep CNN	3D Radar Cavity: 900 high-resolution B-scan images (2019); Cavity: 7,371 Pipe: 7,917 Manhole: 4,914 Intact: 2,730	Confusion matrix (precision, recall, F1-score)	The proposed CNN-based UcNet has incorporated phase analysis of super-resolution GPR images. Misclassification is remarkably reduced compared with conventional CNN models.
Underground voids	Qin and Huang (2016)	SVM with DCT-composed feature vector	FIT simulated GPR images: Voids: 600 scenarios	Confusion matrix	SVM with feature vectors composed of DCT and polarization attributes was applied to the FIT simulation GPR model.
Underground cavity	Ni et al. (2020)	YOLO v3	2D ground-coupled antennas: 270 and 400 MHz Cavity: 2,323 B-scan images	Confusion matrix	YOLO v3 algorithm with a relatively large amount of B-scan with field cavity.

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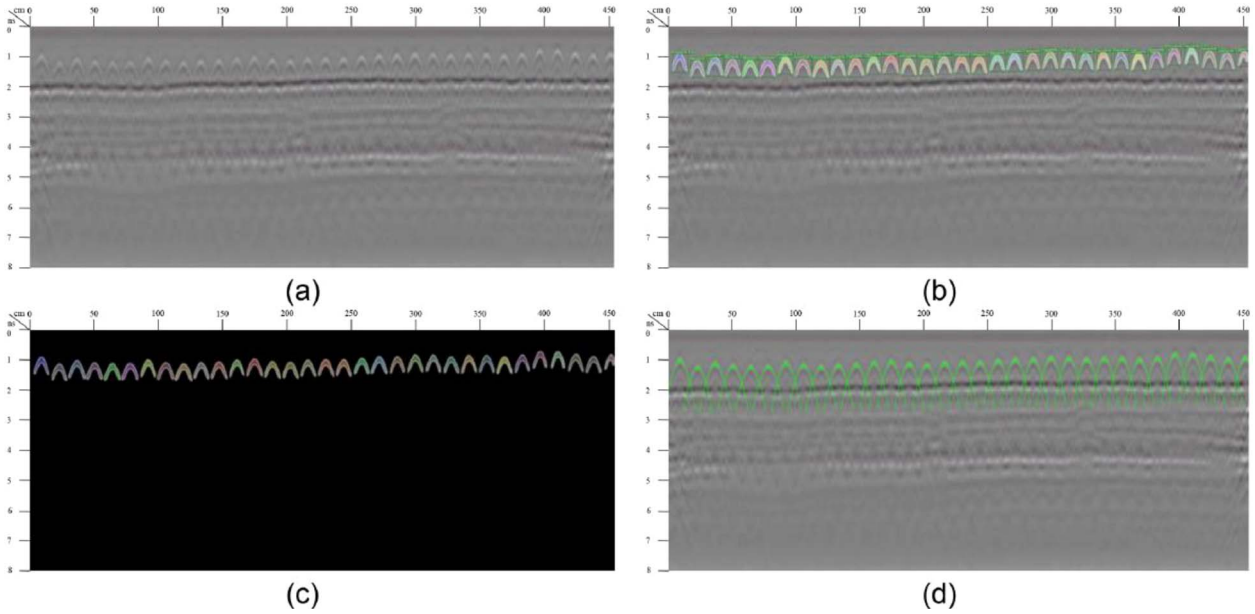


Figure 9. An example of rebar locating using Mask R-CNN algorithm: (a) original GPR scans; (b) segmentation results with predicted box, confidence and mask patch obtained by the enhanced Mask R-CNN; (c) whole mask image; (d) hyperbola fitting results (Hou et al., 2021).

this model on rebar detection and corrosion was verified through a publicly available GPR image dataset called DECKGPRHv1.0. The HOG/AdaBoost model achieved high rebar detection accuracy despite the complexity of GPR field data. In their algorithm, the performance of rebar corrosion detection was affected by various training parameters, such as cell size, block size, block overlap, and the number of orientation bins. It should be noted that the superiority of HOG in hyperbolic feature extraction was reported by other documents as well (Kaur et al., 2015; Noreen and Khan., 2017). Except for the ML-based model, a two-step deep CNN model proposed by Dinh et al. (2018) also achieved excellent performance in automatically locating rebars in the bridge GPR data. First, image processing methods, including migration, normalized cross-correlation, and thresholding, were applied. The pixels that could be near the rebar were located, and the areas were pre-selected. Then, these areas were classified by a trained CNN. Some other DL-based algorithms, such as the improved mask R-CNN with distance-guided intersection over union (DGIoU) (Hou et al., 2021), were also applied, as seen in Fig. 9. Note that the rebar reflection in GPR B-scan is clear and distinct with high accuracy (99.6 per cent for the entire bridge deck in Dinh's work).

3.3.2 *Reinforcement corrosion*

Reinforcement/rebar corrosion evaluation is essential in bridge deck inspection, which could be accomplished using GPR. In the initiation stage of chloride-induced corrosion, the moisture and chloride ions in the surrounding environment intrude into the concrete structure (Hunkeler, 2005; Zaki et al., 2018). Then, hydrated rust products that are volumetrically expansive (four to six times larger) generate during the electro-chemical process (Liu, 1996). Increasing volume leads the reinforcement to break from adjacent concrete. It initiates the formation of micro-cracks in surrounding concrete, which will further develop into macro-cracking, debonding, delamination, and spalling (Lai et al., 2018). This type of deterioration jeopardizes the loading-bearing capacity of the reinforced concrete deck and structural integrity. Evaluation of early-stage corrosion benefits the bridge structure in terms of maintenance/rehabilitation, especially for those in coastal or contaminated areas with rich chloride ions (Vu and Stewart, 2000; Kassir and Ghosn, 2002). Traditionally, two approaches were adopted to evaluate chloride-induced corrosion using GPR quantitatively. The first approach is to establish empirical or theoretical mixing models representing the relationship between bulk dielectric constant variation and volumetric changes before and after corrosion (Halabe et al., 1993; Senin and Hamid, 2016). This approach derives from EM mixing theory in soil mechanics, considering that both soil and cement concrete belong to a porous medium (Topp et al., 1980). The second approach is the full waveform inversion, which maps chloride content, and moisture distribution by inverse GPR attributes such as amplitude, velocity, and dielectric constant (Kalogeropoulos et al., 2011, 2013).

ML- and DL-based methods are recently applied in reinforcement corrosion evaluation using GPR B-scan or C-scan images. The blur of hyperbola reflection amplitudes was proved to have a significant correlation to corrosion conditions near the rebar in concrete bridge deck (Gucunski et al., 2014; Hasan and Yazdani, 2016; Martino et al., 2016). Zhang et al. (2022) propose a three-step single-shot multibox detector (SSD) model to detect rebar corrosion conditions automatically. The proposed model first automatically picked up the rebar location by identifying a bright cluster in a binarized B-scan image. Then, rebar information was extracted to draw the contour map of concrete corrosion, as seen in Fig. 10. The model was trained and tested using 12,894 B-scans in total. The scale of the used dataset is comparatively large. When training the model, images are divided into four categories depending on the hyperbola's clarity and whether they are overlapped. An average of 98 per cent accuracy could be achieved, comparable to in-house software such as RADAN 7 (GSSI, 2023).

ML- and DL-based methods were utilized to replace traditional signal denoising and image enhancement algorithms, aiming at more accurate rebar detection from GPR bridge deck images. For instance, denoising convolutional neural networks (DnCNN) and convolution-denoising automatic encoders were proposed for GPR data denoising (Liu et al., 2018; Si et al., 2019). The hyperbolic feature of rebar in GPR B-scans could be a clutter to other targeted

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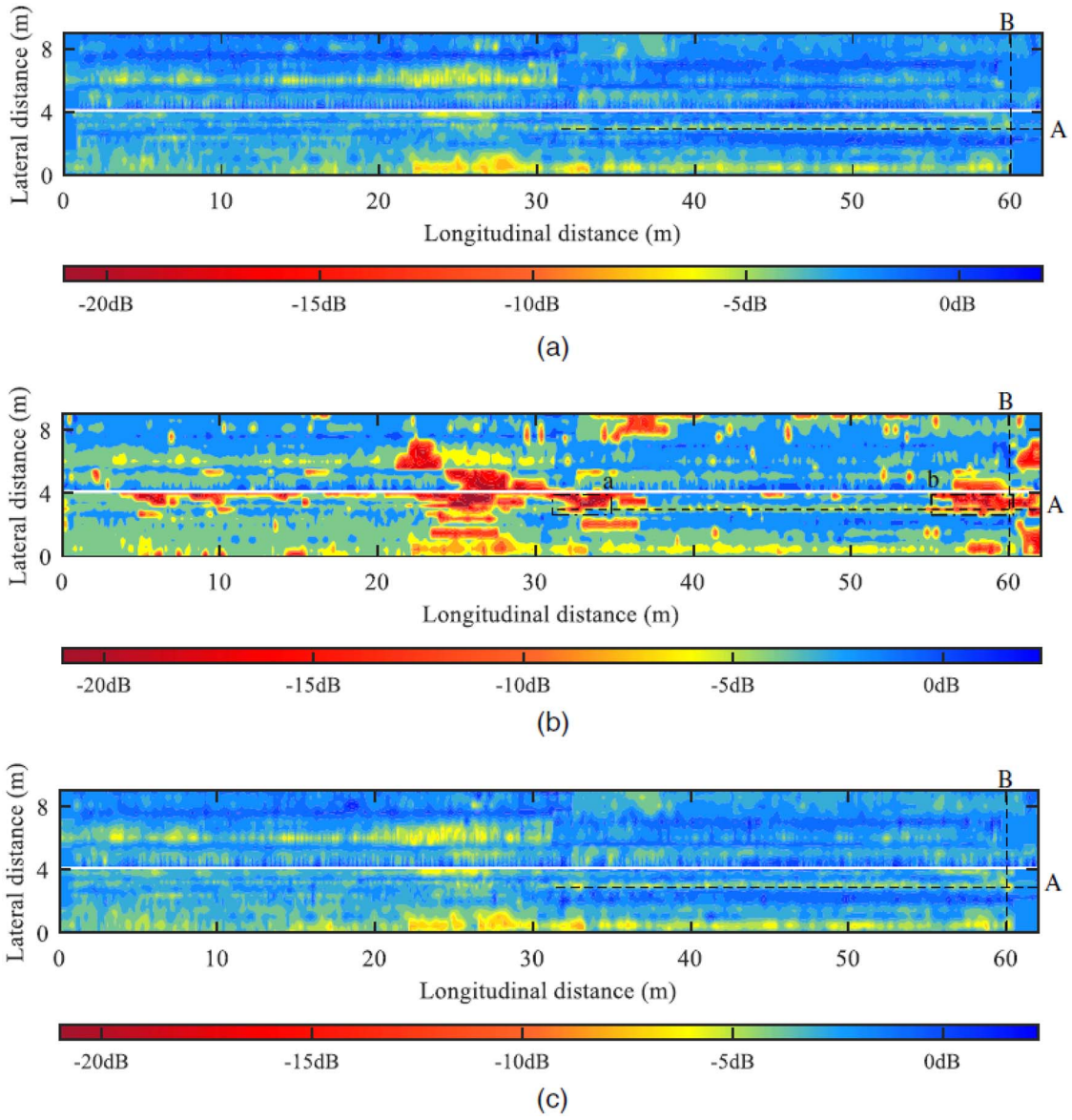


Figure 10. Corrosive environment detection results of reinforcement concrete bridge deck using GPR and DL method: (a) SSD model; (b) commercial software method; (c) manual method (Zhang et al., 2022).

defects with weaker reflection echoes. To address this issue, Wang et al. (2021) proposed a DL-based method whose main characteristics are rebar clutters removal and defect echoes enhancement. In the proposed deep neural network architecture, the residual-inception (RI) block and attention module were modified and added. RI blocks were applied to the multi-branch architecture to identify the regularly distributed hyperbolic feature of rebars to fit the clutter removal purpose. Attention modules were added in convolutional blocks to enhance the feature extraction of the weak defect. The proposed model showed a higher accuracy (0.850) in identifying defects under rebars than the mainstream Faster R-CNN model (0.208) using both synthetic data from FDTD simulation and real data from field application.

3.3.3 *Moisture ingress*

Existing literature shows that moisture is one of the key environmental factors affecting bridge decks' long-term performance (Karbhari et al., 2003; Heshmati et al., 2015). Moisture can be in the form of humidity, liquid water, or de-icing salt. In addition to moisture damage in the asphalt overlay, moisture ingress may occur due to the weakening of adhesively bonded joints and reinforcement corrosion in the delamination areas. The penetration of moisture leads to the formation of micro-cracks, further accelerating the chemical degradation of concrete, spalling, cracking, debonding, and delamination (Karbhari and Ghosh, 2009). Due to the EM wave's sensitivity to moisture in a porous medium, GPR shows the potential to identify regions with different moisture contents on the bridges. However, limited by survey procedure and convenience, recent GPR applications mainly focus on the condition assessment of the asphalt overlay. A dataset containing >8,000 GPR traces with moisture damage on concrete bridge decks was built by Zhang *et al.* (Zhang et al., 2020b). A 28-element time-frequency feature vector was extracted for moisture damage identification, and the element number of the feature vector was reduced to 11 using linear discriminant analysis (LDA). The principal components were decomposed from the processed feature vector using principal component analysis (PCA), which was used

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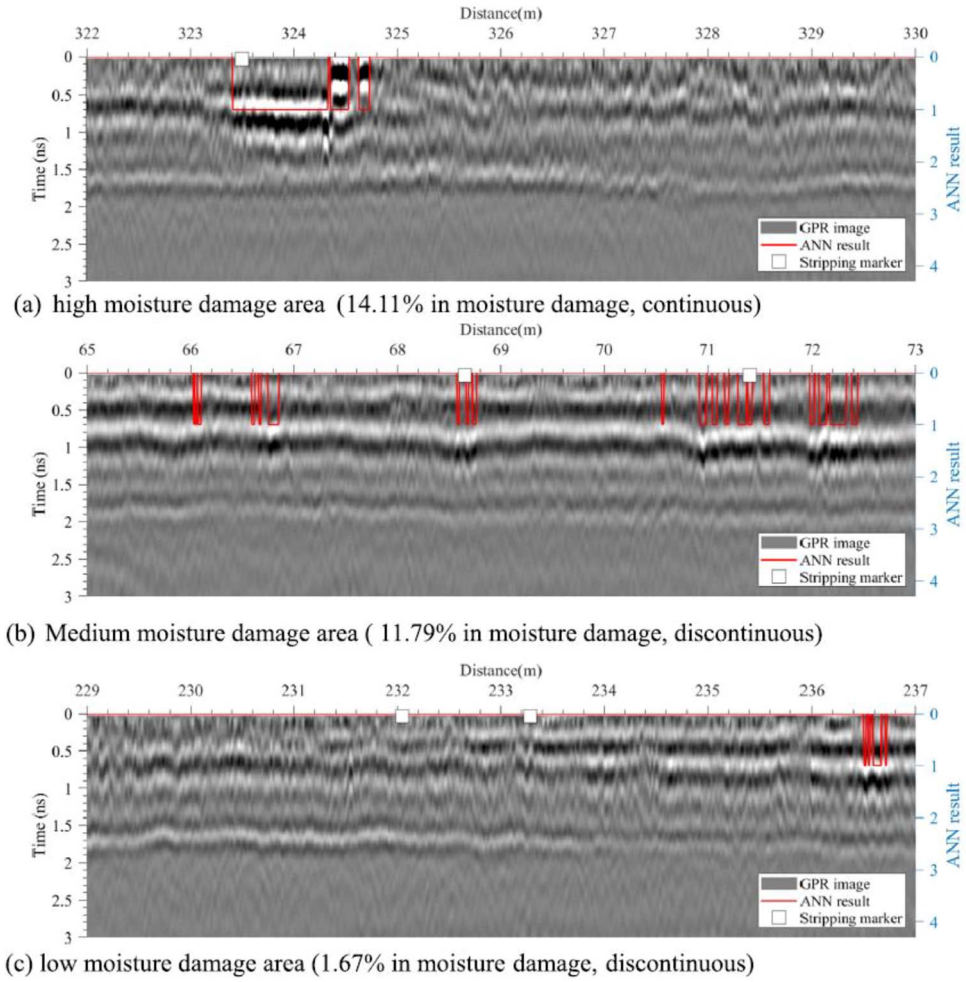


Figure 11. Moisture detection results at different damage degrees using GPR B-scan image and ANN: (a) high moisture damage; (b) medium moisture damage; (c) low moisture damage (Zhang et al., 2020b).

to train a BP-ANN model. Although an accuracy of 92.4 per cent was achieved in terms of moisture damage, the ground-coupled antennas restricted its generalization to data collected by other GPR types, such as the air-coupled antennas.

3.3.4 *Concrete condition*

Attempts have been made to use ML/DL-based algorithms to evaluate concrete conditions. A contour map is preferred to show the health condition of the concrete bridge deck in a bird-eye view. Assisted by an ML-based algorithm, a comprehensive full-depth evaluation of concrete bridge decks could be achieved (Imani et al., 2019).

3.3.5 *Summary*

In summary, ML/DL-based algorithms were utilized to interpret GPR data collected from concrete/reinforced concrete bridge deck defects. Mature application scenarios mainly focus on the identification, locating, and corrosion mapping (deterioration level) of reinforcement bars and moisture ingress in the asphalt overlay. HOG feature descriptors, ANN, CNN, and their variant algorithms have been implemented. However, evaluating cracking, debonding, delamination, spalling, and other common defects of concrete bridge decks using GPR data with ML/DL-based algorithm needs to be improved. A summarization of current ML and DL applications in concrete/ reinforced concrete bridge deck defect detection using GPR data can be seen in Table 2.

4. Existing problems and future work

4.1 Existing problems

Advanced ML/DL-based models outperform traditional velocity analysis methods or signal/image processing-based methods in defect detection of transportation infrastructure using GPR. However, several obstacles are yet to be addressed before they can be extensively applied in practice.

4.1.1 *Scale and quality of GPR labeled data*

Supervised learning is dominant in ML and DL algorithms that have been applied in detecting transportation infrastructure internal defects using GPR. The prediction accuracy requires a large scale of labeled data from synthetic models using numerical simulation or field data. The preparation of large-scale training datasets is time-consuming and labor-intensive. Precise labeling of different defects from intricate GPR images is challenging and requires skillful and experienced operators, especially when encountering field data with multiple noise sources. In addition, manual labeling can be subjective because it is challenging to validate the true subsurface conditions from the data labels. Agreements that regulate typical characteristics of internal defect types in GPR radargram still need to be improved.

4.1.2 *Shared GPR dataset and platform*

Open-source GPR datasets that cover different kinds of internal defects in transportation infrastructure are still lacking.

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Table 2. ML/DL applications upon concrete/reinforced concrete bridge deck defect detection using GPR data.

Tasks	References	Algorithms and methods	Data source and scale	Performance metrics	Features
Rebar detection	Asadi et al. (2019) Asadi et al. (2020)	HOG, AdaBoost, MLP	Ground-coupled 2D radar B-scan:4,000 images	Confusion matrix (precision, recall, ROC, F1-score)	Computer-aided rebar detection chain consists of a fined-tuned HOG-/MLP-based binary image classifier.
Rebar detection	Dinh et al. (2018)	CNN	Ground-coupled 2D radar B-scan: 4,000 images	Confusion matrix (differently defined accuracy)	A two-step deep CNN model was developed
Rebar clutter removal and defect echo enhancement	Wang et al. (2021)	DNN, RI block, attention	Synthetic data (FDTD): 72,000 B-scan Actual data: 12,000 groups after data augmentation	Peak PSNR Mean structural similarity Confusion matrix	A deep neural network model characterized by added RI block and attention module was proposed for the purpose of rebar clutter removal and defects echo enhancement.
Rebar detection	Hou et al. (2021)	Improved Mask R-CNN, DGIoU	Ground-coupled palm radar, 95 B-scan	AP	An improved mask R-CNN model incorporated by DGIoU was proposed to detect and segment rebar in concrete bridge decks simultaneously.
Rebar corrosion detection	Zhang et al. (2022)	SSD	1.5 GHz ground-coupled antenna 12,894 B-scan images	Accuracy	A three-step SSD model was built. The corrosion condition of the bridge deck was mapped by using the Hyperbolic features of rebar.
Moisture ingress in asphalt overlay	Zhang et al. (2020b)	PCA, BP ANN	Ground-coupled radar, 8,215 traces of moisture damage	Accuracy	LDA was used to optimize the extracted feature vector; PCA was used to decompose the feature vector. BP-ANN model was trained.
Rebar deterioration detection	Kaur et al. (2015)	SVM and clustering and curve fitting	IDS and GSSI ground-coupled radar, 3,000 samples extracted from B-scan	Accuracy, precision, sensitivity, specificity	The SVM method was first used to localize the position of the rebar. Then, hyperbola region pruning was fulfilled by clustering, followed by hyperbola fitting for deterioration mapping.
Concrete condition	Imani et al. (2019)	ML: gradient boosting	GSSI 1.5 GHz radar, 30,696 hyperbolas for the top layer; 30,000 hyperbolas for the middle layer	Accuracy	An ML-based algorithm, namely gradient boosting, was used to evaluate and rate the concrete condition within and beyond the top reinforcement zoon to fulfill a full-depth assessment of the bridge deck.

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Repetitive and similar workflows followed by different academic research groups lead to redundant contributions and waste of computing resources. In addition, raw data formats, collection modes, and coupled processing tools may vary from different GPR suppliers. This also leads to the incompatibility of GPR data from different vendors, which forces practitioners and researchers to repeat tasks using their own equipment and databases at low efficiency.

4.1.3 *Customized architecture and algorithm generalization*

GPR images require customized methods to recognize and detect specific patterns generated by subsurface targets in transportation infrastructure (Hou et al., 2022b). Subsurface targets are often identified by extracting hyperbola from rippled and blurred radargrams due to the dielectric complexity of the background medium. Meanwhile, as the 3D GPR system becomes more prevalent, the synthetic use of GPR B-/C-/D-scan data contributes to the reconstruction of a complete morphology of the internal defects. To achieve this goal, DL architectures that fit small object detection and match unique hyperbola features are required to be modified. In-depth workflows that integrate advanced signal and image processing methods and cutting-edge ML and DL approaches could be further optimized. Besides the lack of customized DL architecture, it is challenging to propose a DL algorithm that can be applied to all defect types in transportation infrastructure.

4.1.4 *Field validation*

The generalization performance of trained ML/DL defect detection models is typically evaluated using the test dataset, which usually shares the same GPR data source as the training dataset. However, the defects position and geometry, which were predicted using ML/DL methods, have been rarely verified by ground truth information. For pavement and bridge overlay, ground truth evidence can be acquired by *in situ* core samples. Drilling cores from pavements or bridges need careful examination by the transportation department to avoid traffic interruption and structural damage. Many DL-based models show superiority and high prediction accuracy in defect detection using a limited scale of GPR data. By contrast, cross-validation by project-level field core data is still limited.

4.2 Recommended future work

To solve the existing issues, three possible future research tasks are summarized as follows: transfer learning technique, integrated robotic platform, and data fusion.

4.2.1 *Transfer learning*

Transfer learning is a technique that could be utilized in specific scenarios to solve the requirements of big data. This technique 'recycles' pre-trained knowledge from one task to different related tasks to reduce reliance on huge dataset. In the GPR community, transfer learning has been verified as efficient in GPR A-scan-based soil water content prediction (Zheng et al., 2019), B-scan-based underground buried wire detection (Aydin and Erdem, 2019), and buried defects classification (Bralich et al., 2017; Reichman et al., 2017; Elsaadouny et al., 2020). The immense potential is under exploration in terms of using transfer learning to reduce the demand for large-scale GPR datasets. However, it should be noted that this technique only learns low representations from benchmark datasets. Limited contributions could be offered in the fine-tuning phase of pre-training (Tong et al., 2020a).

4.2.2 *Integrated robotic platform*

Portable GPR systems carried by UAV platforms have drawn attention in the near-surface remote sensing communities. Reduced vibration effect during measurement and flexible survey routine facilitates its blooming implementations. For example, a helicopter-mounted GPR system has been successfully applied to solve land and geophysical survey tasks (Forte et al., 2019). Drone-mounted 3D GPR array has been used in pavement density prediction (Nordic Unmanned, 2021). Before being widely applied, the low payload and limited battery capacity need to be solved. Besides UAV, other robotic platforms that could simultaneously integrate multiple NDT methods have been established in many civil engineering scenarios, including the robotics-assisted bridge inspection tool integrating GPR, digital camera, IR, and GPS for bridge deck inspection (Gucunski et al., 2017) and robotic system for road and airport runway inspections (Gui and Li, 2020).

4.2.3 *Data fusion*

Data fusion from multiple NDT measurements using ML- and DL-based strategies may overcome the limitations of using GPR alone. For instance, Li et al. (2022b) adopted YOLO v3 and 1D CNN for reinforcement rebar recognition with combined GPR and electromagnetic induction (EMI) data. Moghadas et al. (2010) used the NB method to interpret fused data from GPR and EMI. Using data acquired by different NDE methods, more accurate structural information could be obtained, such as the cover thickness and diameter of reinforcement bars and hyperbolic feature extractions. GPR data fused with IR, falling weight deflectometer (FWD), and intelligent compaction (IC) under the ML/DL framework may provide high internal defect detection accuracy in transportation infrastructure.

5. Conclusions

A comprehensive review of the applications of ML and DL algorithms in detecting the internal defects of transportation infrastructure (pavement and bridge) using GPR is presented in this paper. The main features of the applied algorithms categorized by defect types are analyzed. This review paper is expected to provide researchers, trainees, and practitioners with the working principles of GPR and ML/DL as well as the status and gaps of ML/DL-based engineering GPR practices. The main conclusions of this review are summarized as follows:

- In road pavement and airport runways, ML/DL methods have been mainly applied to identify internal defects such as concealed cracks, underground cavities and voids, moisture damage, subgrade defects, uneven settlement, manhole, and joint covers. CNN-based object detectors using GPR B-scan, such as multi-stage CNN, NIN, R-CNN, and YOLO series, show better performances than other alternatives. B-scan images collected by the 2D GPR system are the mainstream raw data formats. Automatic DL-based defect prediction models integrating both B-scan and C-scan show promising accuracy, which could accelerate the development and application of 3D radar systems.
- In bridge applications, ML/DL-based methods have been mainly applied to locate and map reinforcement/rebar and evaluate corrosion degrees. Detection of moisture ingress through asphalt overlay is an essential application scenario in bridge deck inspection. The ML-based algorithms, including HOG, AdaBoost, MLP, PCA, BP neural network, ANN, and SSD,

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as well as the DL-based algorithms, including DNN, Mask R-CNN, CNN, Faster R-CNN, and DnCNN, have been applied. An intuitional assessment for structural health monitoring can be achieved via contour maps of the concrete condition of the bridge deck using the ML and DL methods in an automatic manner.

- The ML and DL applications in detecting defects of transportation infrastructure using GPR rely on a large number of high quality labeled images. The lack of shared databases of different internal defect types leads to redundant attempts and endeavors by different academic research groups in the GPR community. The lack of customized DL architecture and poor transfer ability hinder its application scenarios. Destructive coring operations make the project-level validation of ML/DL model prediction accuracy challenging. Transfer learning techniques may significantly reduce the training dataset requirements. Integrated robotic platforms, such as UAV and automatic inspection robots, as well as the data fusion between GPR and other NDT methods, such as EMI, IR, FWD, and IC, could accelerate GPR applications in transportation infrastructure.

Conflict of interest statement

One of the authors, Dr. Zhen Leng, is a member of editorial board of Intelligent Transportation Infrastructure. He was blinded from reviewing or making decisions on the manuscript.

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